

Sanchayan Bhowal

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EDUCATION

Stanford University PhD in Statistics	Stanford, CA 2025-Present
Indian Statistical Institute Master of Mathematics, Grade: 97.7% <i>Advisors:</i> Dr. Bhaswar Bhattacharyya and Dr. Somabha Mukherjee Bachelor of Mathematics, Grade: 94.5%	Bangalore, India 2023-2025 2020-2023

RESEARCH

- Bhattacharyya, B., **Bhowal, S.**, Sengupta, A. (2026). Thresholds and fluctuations for colorful arithmetic progressions in sparse random colorings 
- **Bhowal, S.**, Chatterjee, A., & Mukherjee, S. (2026). Scaling Limits for Ising Models on Inhomogeneous Random Graphs and Applications 
- Mukherjee, S., **Bhowal, S.**, Chatterjee, A., & Bhattacharyya, B. (2026). Ising Models on Inhomogeneous Random Graphs: Inference, Local Asymptotic Minimality, and Limit of Experiments 
- Bhattacharyya, B., **Bhowal, S.**, Das, K., Eslava, L. & Karmakar, S. Thresholds and Fluctuations in Random Multiplex Networks. *Canadian Mathematical Bulletin* 
- **Bhowal, S.** & Mukherjee, S. Rates of Convergence of the Magnetization in the Tensor Curie-Weiss Potts Model. *Journal of Statistical Physics* 192, 2 (2025). 
- **Bhowal, S.** & Mukherjee, S. (2023). Limit Theorems and Phase Transitions in the Tensor Curie-Weiss Potts Model. *Information and Inference: A Journal of IMA*. 
- **Bhowal, S.**, Samanta, R., Ray, A., Bhattacharyya, S. & Hens, C. (2023) Exploring the potential of collective learning to reduce foraging time. *Chaos, Solitons & Fractals*, 168, 113123 

EXPERIENCE

Big Data Summer Institute, University of Michigan Dr. Snigdha Panigrahi & Dr. Johann Gagnon-Bartsch	June 2023 – July 2023
SN Bhatt Memorial Fellowship Program Dr. Anirban Basak & Dr. Riddhipratim Basu	May 2022-July 2022

TALKS

Triviality of Ising models in Inhomogeneous Graphs Invited talk at Bangalore Probability Seminar	2026
Limit Theorems of Estimators in The Tensor Curie Weiss Potts Model Invited talk at Bangalore Probability Seminar	2024
What is Cutoff? Math Club ISI Bangalore	2024
A tour of Markov chains Math Club ISI Bangalore	2022
An Unconventional Approach to Groups ISI Students Seminar	2021

AWARDS AND FELLOWSHIPS

- P.C. Panesar Gold Medal for outstanding performance in M. Math Programme (2023-25)
- Regional prize Madhava Mathematics Competition (2023)
- National award ISLP Poster Competition (2022)
- KVPY fellowship (2020- 2025)
- JBNSTS Scholarship (2018-2020)

TEACHING EXPERIENCE

STATS 200: Introduction to Statistical Inference	Fall 2026-27
STATS 60: Introduction to Statistical Methods: Precalculus	Summer 2025-26
STATS 60: Introduction to Statistical Methods: Precalculus	Winter 2025-26
STATS 110: Introduction to Statistics for Engineering and the Sciences	Fall 2025-26

CONFERENCES

Geometry, Probability, and Algorithms Organizers: Akash Kumar, Anand Louis and Piyush Srivastava	May 12-16 2025
High Dimensional Expanders and Error Correcting Codes Organizers: Irit Dinur, Venkat Guruswami and Prahladh Harsha	May, 1-9 2025
Mathematical Foundations of Network Models and their Applications Organizers: Louigi Berry, Siva Athreya, Shankar Bhamidi, Serte Donderwinkel, Soumik Pal	Dec, 9-21 2024
International Colloquium on Randomness, Geometry, and Dynamics Organizers: TIFR Mumbai and IISER Pune	Jan, 1-5 2024
Topics in High Dimensional Probability Organizers: Anirban Basak and Riddhipratim Basu	Jan, 2-13 2023
Vijyoshi Organizer: IISc Bangalore for KVPY fellows	Dec, 7-9 2019

SKILLS

Programming Languages: C/C++, Python, R, JavaScript, HTML/CSS, $\text{\LaTeX}$
Languages: Bengali(Native), English, Hindi, Assamese

PROFESSIONAL SERVICES

Reviewer: Annals of Applied Probability, Journal of Physics A: Mathematical and Theoretical, Sankhya B: The Indian Journal of Statistics.

Organizational Roles

- Student Coordinator, **Math Club** ISI Bangalore, 2022-2025
- Core Team Member, LIMIT, an online mathematics competition, 2021-2025